

Section B

10 (a)	light-dependent resistor (allow LDR)	B1	[1]
(b) (i)	two resistors in series between +5 V line and earth midpoint connected to inverting input of op-amp	M1 A1	[2]
(ii)	relay coil between diode and earth switch between lamp and earth	M1 A1	[2]
(c) (i)	switch on/off mains supply using a low voltage/current output (allow 'isolates circuit from mains supply')	B1	[1]
(ii)	relay will switch on for one polarity of output (voltage) switches on when output (voltage) is negative	C1 A1	[2]
11 (a) (i)	e.m. radiation produced whenever charged particle is accelerated	M1	